

SMART DIE : Final-Face Prediction and Motion Reconstruction with Self-Attention

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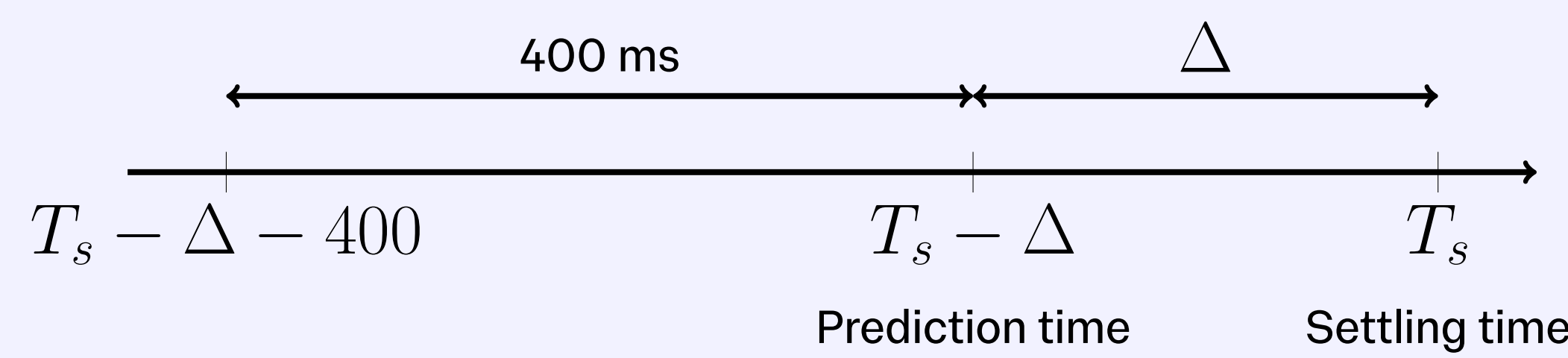
Motivation

An ordinary six-sided die provides information only about its final outcome. It records no information about its motion leading up to the settling of the final face. This project aims to collect time-series data from a rolling die, train and use classification methods to investigate how early in a roll its final outcome becomes predictable. Additionally, we explore the motion reconstruction abilities of a self-attention neural network.

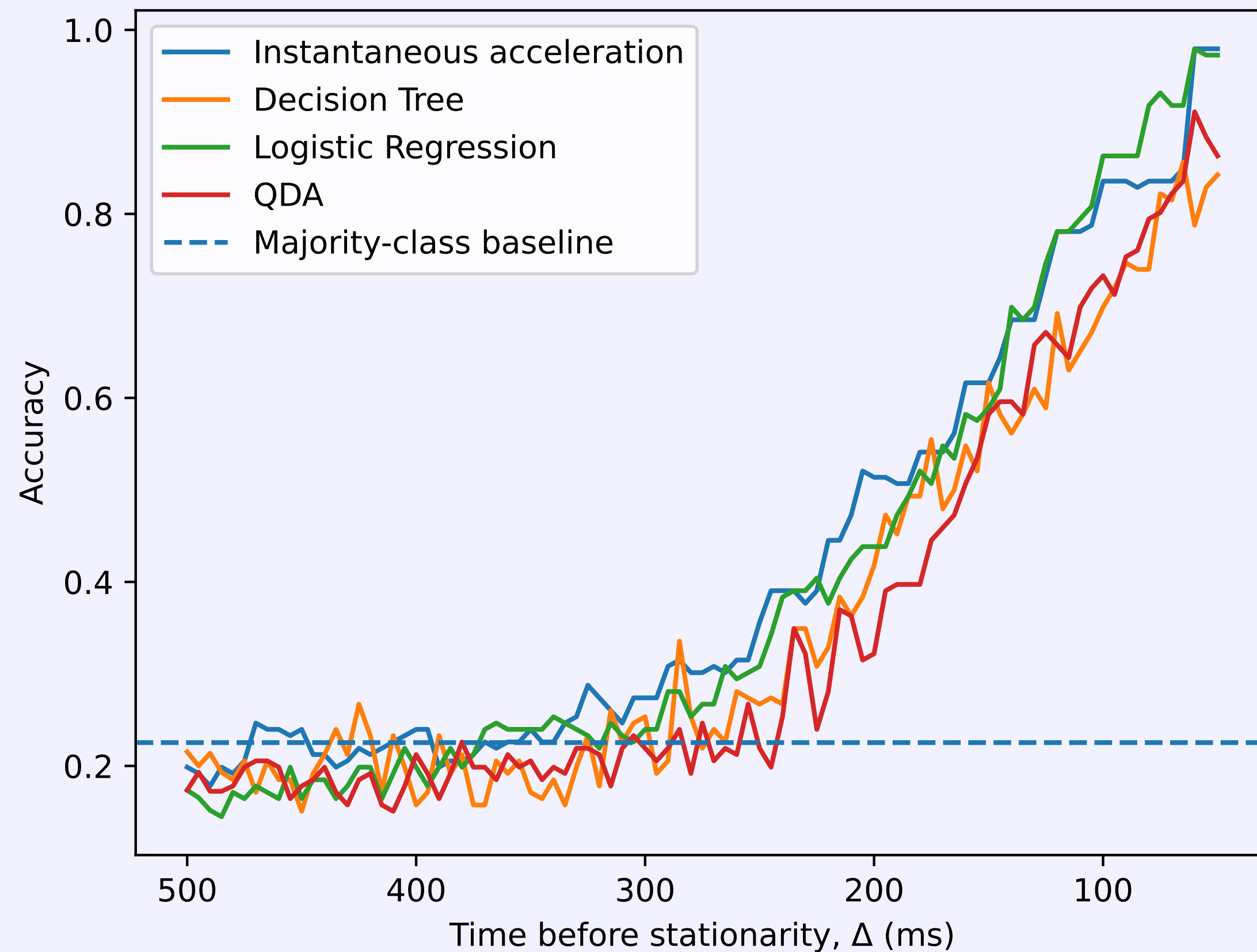
Final-Face Prediction

The IMU takes measurements of the form $x(t) \in \mathbb{R}^6$, where

$$x(t) = (a_x(t), a_y(t), a_z(t), \omega_x(t), \omega_y(t), \omega_z(t))$$



The classification models [1] received 400 ms of IMU data, ending Δ ms before the detected settling time T_s . Each window was resampled to 20 data points, producing a 20×6 matrix, which was flattened to a 120-dimensional feature vector.

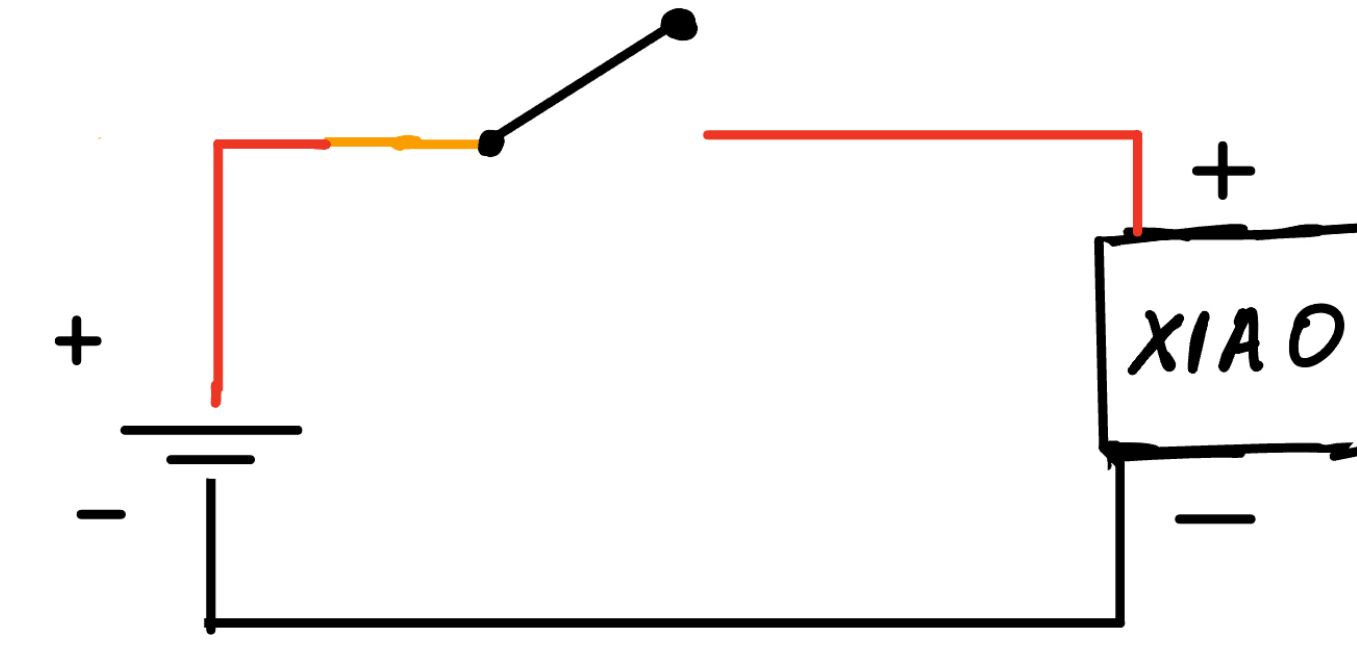
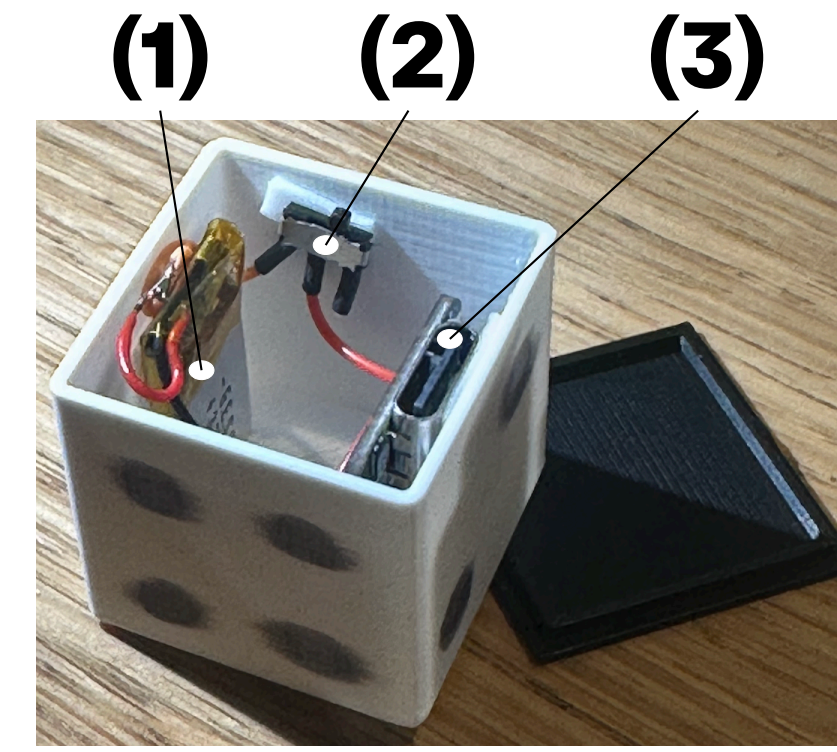


Prediction from instantaneous acceleration is given by

$\hat{F} = \arg \max \{-a_x, a_z, a_y, -a_y, -a_z, a_x\}$, the ordering corresponds to $1, \dots, 6$.

Instantaneous acceleration performs similarly to models using the 400 ms history, suggesting that information about the final face is already present in the current acceleration vector.

Part Labels, Measurements and Circuit



(1) 3.7 V LiPo battery (2) Switch (3) XIAO nRF52840 Sense

The final die measurements are 26 mm $\times$ 25.8 mm $\times$ 25.6 mm. The shell and lid were designed in Fusion and 3D printed. The battery, switch and microcontroller were assembled and soldered by hand.

Motion Sequence Reconstruction (MSR)

Here we investigate whether the original motion sequence of a roll can be recovered from the IMU measurements and final face. The final 300 ms before settling is resampled into 15 states, $x_1, \dots, x_{15}$.

No positional encoding is used, since the input order has been randomly shuffled by a permutation π . Each state is assigned its relative position, $\tau_i = \frac{i-1}{14}$ $i = 1, \dots, 15$. Importantly, the network uses

$$\hat{a} = \frac{a}{\|a\|} \quad \text{and} \quad \hat{\omega} = \frac{\omega}{\|\omega\|}$$

which means it must rely on patterns in directional change, rather than the magnitude of these vectors. **Self-Attention:** Each state is mapped to a learned embedding and combined with an embedding of the final face, F . The resulting representations form matrix $X \in \mathcal{M}_{15,32}(\mathbb{R})$, which is transformed into **query**, **key**, and **value** matrices Q, K and V :

$$Q = XW_Q \quad K = XW_K \quad V = XW_V$$

$$Attention(Q, K, V) = softmax \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad [2]$$

QK^T scores the relationship between every pair of states, allowing the network to learn relationships between measurements and infer their relative order. $\hat{\tau}_i$ prediction flow diagram:

$(x_{\pi(i)}, F)$ Embedding $\rightarrow$ Multi-Head Attention $\rightarrow$ Feed-Forward $\rightarrow \hat{\tau}_i$

Network is trained by minimising

$$\mathcal{L}_{MSE} = \frac{1}{15} \sum_{i=1}^{15} (\hat{\tau}_i - \tau_i)^2$$

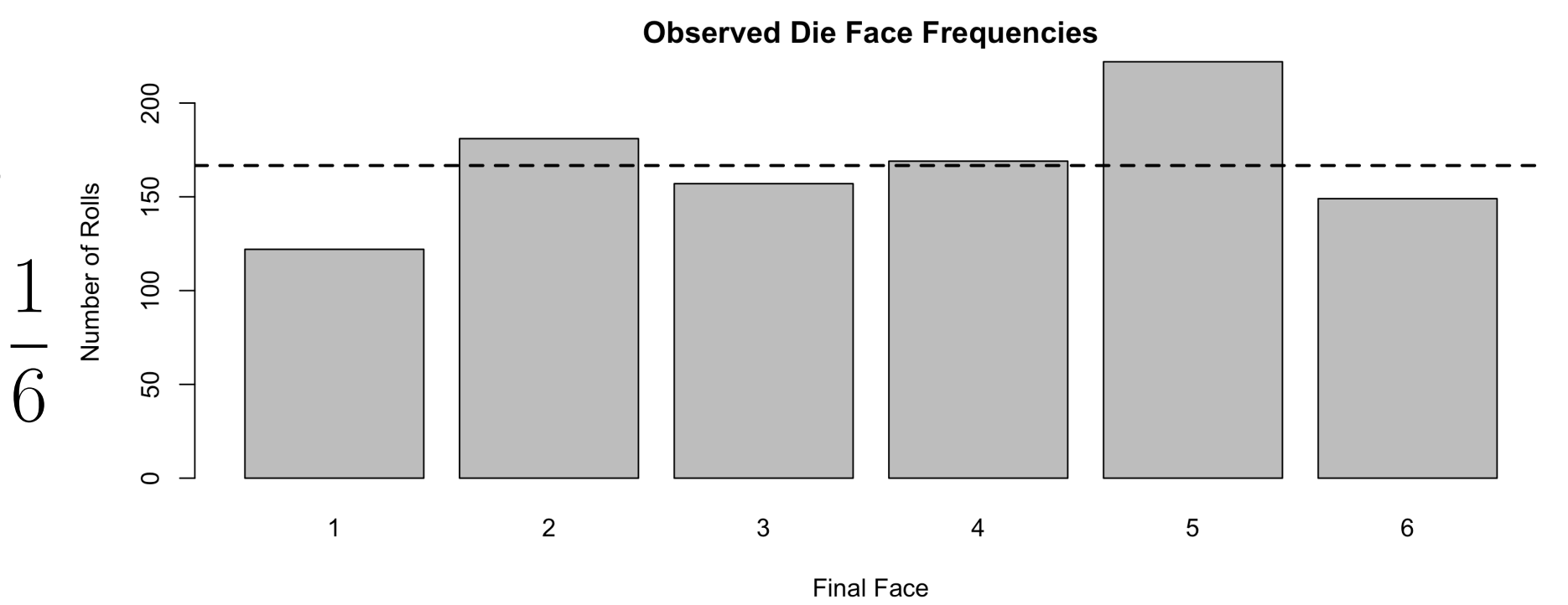
Sorting the states by $\hat{\tau}_i$ gives us the reconstructed motion sequence.

Bias Hypothesis Test

Due to the weight distribution of the components in the shell, we test whether the die is fair. $X_i :=$ the number of times face i occurs. After n rolls, $(X_1, \dots, X_6) \sim \text{Multinomial}(n; p_1, \dots, p_6)$.

$$H_0 : p_1 = \dots = p_6 = \frac{1}{6}$$

$$H_1 : \text{at least one } p_i \neq \frac{1}{6}$$



Using a χ^2 goodness-of-fit test, we obtain a p -value of 2.3×10^{-6} , providing strong evidence against H_0 and suggesting the die is biased. Our baseline for prediction accuracy is

$$\hat{A} = \max_k \hat{p}_k$$

where $\hat{p}_k$ are the empirical probabilities calculated from the training data. **Dataset:** 1,000 physical rolls were recorded. Of these, 970 rolls contained enough pre-settling data for the prediction and sequence-reconstruction.

MSR Results

We measure reconstruction performance using pairwise ordering accuracy: the proportion of the $\binom{15}{2} = 105$ pairs for which the network correctly identifies which state came first. We also calculate the Spearman rank correlation ρ_s between the true and predicted positions. **The results on the held-out test set (146 rolls) are:**

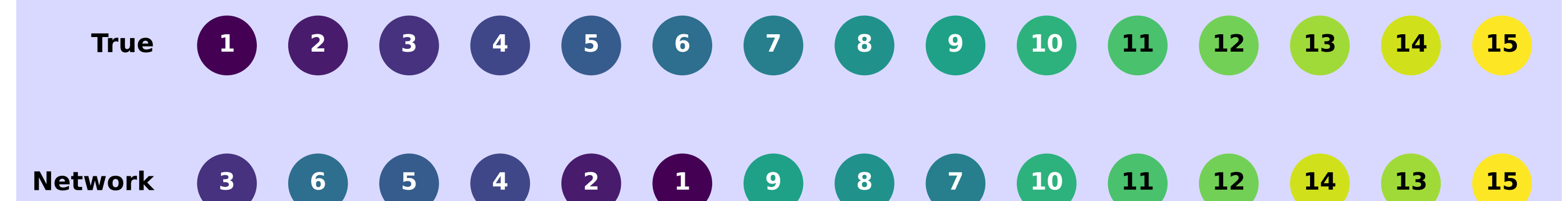
Random baseline pairwise accuracy : 50%

Mean pairwise accuracy = 81.8% (95% CI : 80.3% - 83.2%)

Mean Spearman $\rho_s = 0.782$ (95% CI : 0.750 - 0.811)

where the confidence intervals were obtained using the bootstrap procedure.

Below is a reconstruction of randomly selected test Roll 857.



These results suggest that directional changes in acceleration and angular velocity contain substantial information about the ordering of states during settling.